

Safely: Platform Design Challenge Report

Deliverable 1: Framing the Problem

Motivation for Choosing Safely

In today's day and age, there exists a very evident safety challenge in certain parts of Europe, especially the south of France and Italy. As of 2023, the crime index in Rome remains to be an evident 53.0 which is far from the ideal crime index of 20.0 and under to classify a region as safe (Numbeo, "*About Crime Indices*"). It is further notable that although crime impacts all to an extent, the most susceptible individuals are on-foot pedestrians, including both - locals and tourists, whom we define as the targeted market segment for our project.

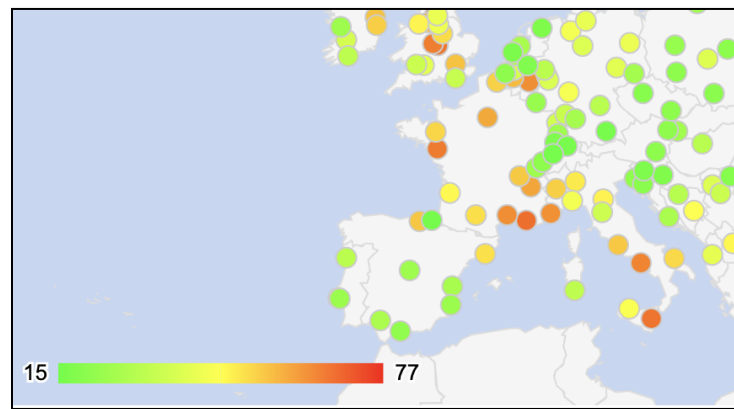


Fig. 1 Crime Index in Europe as a Heatmap (Numbeo, "*Europe: Crime Index by City 2023*")

With Safely, a personal travel safety guide constructed into a convenient app, our idea is to facilitate a safer environment for frequent local and tourist walkers throughout the streets of all urban cities. Our decision to develop Safely is rooted in a genuine concern for the safety of people navigating the urban streets as we believe that every pedestrian should have the freedom to move around the city confidently. Individual encounters with safety issues have driven us to seek a solution that enables individuals to make informed choices about their routes and safety. Further, our personal experiences make us better understand what features in a safety app would best facilitate a secure environment.

Context and Safely as a Platform Business

Safely, as a platform business, offers a local solution to Italy as a country with the potential for a global reach in the urban planning and smart city solutions industry, extending peripherally into the tourism industry by harnessing modern technology and community generated data. Safely empowers users to make informed decisions about their routes, fostering a sense of security and confidence. With Safely, we also aim to unite two distinct user groups – pedestrians and local entities – to collectively enhance urban safety.

Further, increased safety and lower crime index would enhance the tourism experience throughout Europe which is a very notable point since as of 2023, the tourism sector in Europe is predicted to contribute €1.44TN to the economy, edging close to the 2019 pre-pandemic high of €1.47TN (World Travel & Tourism Council). Therefore, our motivation for this project stems from an altruistic attitude towards safety and aiding to increase economic benefit through safer tourism.

Value Proposition

The defined problem formerly becomes an increasing concern for not only local residents, but also tourists who might not engage in active tourism activities due to safety concerns, leading to severe short-term and long-term problems. While the immediate harm is physical (and sometimes financial) damage due to crime (theft, violence and assault), the extended problems come in the form of decreased tourism, a primary source of Italy's (and Europe's) economy and possible migration away from high crime index cities. Safely's value proposition focuses on enhancing personal safety for both residents and tourists. Safely empowers individuals to confidently explore cities, ensuring their journeys are not only enjoyable but also more safe. It not only aims to reduce the immediate risks of physical and financial harm due to crime but also contributes to mitigating the broader economic impact.

Deliverable 2: MultiSidedness and Network Effects

The multi-sidedness of our platform is moderately diverse as we facilitate a heatmap of the user's city where red indicates very high crime index areas and blue indicates very low crime index.

We specify this feature for our premium users by introducing a GPS with real time tracking and a thorough analysis of whether the user's selected route (the path from start destination to end destination) is classified as safe or not as per the respective crime index of the region in interest.

This allows us to make the app accessible to users who might not have an immediate incentive to pay whilst allowing us to provide premium services for users that are willing to pay. Here, we notice that the nature of our core interactions is primarily Consumption, that is, in every interaction there is, at least, one user who consumes value (value referring to the heatmap of the crime index). We provide further the report section which allows users to report a crime if they encounter it by themselves which also enables Creation, that is, in every interaction there is, at least, one user who creates value (value being reporting of crime).

User Groups and Interactions:

Primary Users: The primary users of Safely are individuals, especially those new to big cities, who seek safe navigation in urban areas. These users benefit from realtime safety information, helping them make informed decisions about their routes and destinations.

Premium Users: The premium users will be given the access to allow location sharing and enable GPS real time tracking to allow an assessment of the safety of their route.

Reporting: A key feature of Safely is the ability for users to report dangerous encounters or safety concerns. This user generated content enriches the platform's safety data, making it more comprehensive and reliable for all users.

SOS Feature: The SOS button is a critical safety tool, providing users with quick access to emergency numbers. This feature enhances user trust and reliability in the platform, especially in urgent situations.

Network Effects in Safely

Direct Network Effects: As more users join and actively use Safely, the platform becomes increasingly valuable. A larger user base leads to more data on safe routes and reported incidents, improving the platform's utility for every user.

Indirect Network Effects: User reports contribute to a richer database of safety information, attracting more users seeking reliable safety insights. This reciprocal relationship between content generation (user reports) and consumption (safety information) strengthens the platform's overall value proposition.

Positive Externalities: Each additional user who reports safety incidents or uses the SOS feature contributes to a safer community. These actions provide direct benefits to the individual user and indirect benefits to the broader Safely community by enhancing overall safety awareness.

Main Externalities: Users also engage in cross-group externalities as the group of users reporting a crime (action) impacts the group of users looking at the heatmap for safety information. Further, Safely operates as a multi sided platform, connecting pedestrians and local entities (businesses and law enforcement). Same group externalities do not exist as the platform provides users data privacy by not sharing information as private as live location with other users.

Conclusively, user groups and the network effects arise from their interactions on the Safely platform. The focus is on how the platform becomes more valuable as the number of users increases and how user contributions (such as safety reports) enhance the platform for all users.

Deliverable 3 : Safely's Platform Business Model

This business model map illustrates the symbiotic relationships among Safely's diverse stakeholders, highlighting how each participant contributes to and benefits from the platform. It underscores the importance of a collaborative approach to urban safety, where every stakeholder plays a vital role.

Customer Type	Value Received from Others	Value Received from Safely	Value Provided to Others	Value Provided to Safely
Pedestrians	Safety tips and route suggestions from other users. Local offers from businesses.	Realtime safety information. Navigation tools. Community support.	Sharing route experiences and safety tips. Feedback on businesses and areas.	User data for safety analytics. Community engagement.
Local Businesses	Customer traffic from pedestrians using Safely.	Advertising platform. Insight into pedestrian traffic patterns.	Local offers and promotions to pedestrians. Support for safer neighborhood initiatives.	Advertising revenue. Partnerships for community safety programs.
Municipal Services (e.g., Law Enforcement)	Community insights for better urban planning.	Data analytics for urban safety. Collaboration platform for community engagement.	Improved safety measures and responses based on user data. Community outreach and safety initiatives.	Data for urban safety planning and response. Enhanced community relations.
App Developers (for additional features)	Platform to showcase and monetize their apps.	Access to a safe user base. Integration support.	Advanced features and apps enhancing user experience.	Innovative features attracting more users. Revenue share from premium features.

Business Model Phases

Safely's business model will evolve in phases, catering to the diverse needs of its user base while ensuring the platform's sustainability:

Initial Phase: Free Access

In its initial stages, Safely will operate as a free-to-use platform. This approach aims to rapidly build a user base by offering essential safety information, such as identifying "safe" areas within a city. This phase focuses on user acquisition and engagement, laying the foundation for a robust and active community.

Transition to Premium Version

As Safely establishes its user base and proves its value, the introduction of a premium version is planned. This premium service will offer enhanced features, including:

Real-Time Tracking: Users can select a walking route from point A to B, with the platform providing real-time tracking for the duration of their journey. This feature ensures continuous safety monitoring and provides peace of mind for users navigating unfamiliar or potentially unsafe areas.

Future GPS Device Integration: Looking ahead, Safely aims to integrate a physical device, such as a GPS-enabled ring, for situations where a phone may not be available or out of charge. This device would offer an additional layer of security and convenience, particularly valuable in emergency scenarios.

Long-Term Monetization Strategy

The transition to a premium model reflects Safely's long-term monetization strategy. While the core functionality remains accessible to all users, premium features provide added value for those seeking enhanced safety tools. This approach balances accessibility with revenue generation, ensuring the platform's sustainability and continuous improvement.

We create a sustainable revenue model that aligns with the users' needs and expectations.

Freemium Model: We can provide essential safety features for free, ensuring basic route suggestions and safety tips are accessible to everyone. For those seeking advanced capabilities, we can offer a premium version with real-time alerts, detailed safety analytics, and personal safety tracking.

Advertisements: We can integrate minimal and non-intrusive advertising within our app, partnering with local businesses that align with our safety-first philosophy, such as reputable taxi services and safe dining establishments.

Data Insights: We can aggregate and anonymize user data to identify and track safety trends, which we can then offer to urban planners and local authorities, aiding them in making informed decisions to improve city safety.

In-App Purchases: We can make one-time services available for in-app purchase, like a safety escort service for users who may need immediate assistance but do not have a subscription.

Crowdsourcing and Community Support: We can create avenues within our app for users to actively contribute to their community's safety, either through donations or by funding specific safety initiatives.

Our monetization plan for Safely is set up to support the app's ongoing operation and development. We can develop a system of free basic features with advanced options that users can pay for. This strategy generates a consistent income, which we reinvest back into the research and development of the app to enhance its capabilities and the overall user experience.

Ethical Consideration

In order to demonstrate a strong commitment to ethical data use and the protection of user privacy, we acknowledge the sensitive nature of location data and personal safety information. The platform adheres to strict data protection standards to ensure the security of user information. We practice basic data integrity methods such as access control and data encryption in order to make sure that there is no unintentional data leakage

Regarding data use and transparency, the platform utilizes user data exclusively for enhancing safety analytics and delivering real-time updates. We ensure that there is a consent policy every time user data is being used for any internal or external research. While sharing data with other organizations, we make sure to hide any unnecessary private information, ensuring anonymity of all users.

Deliverable 4: Testing and Prototyping (One Page)

Testing and Validation Strategy

Utilizing Government Data:

We can harness extensive datasets from municipal and government sources to establish the foundation of Safely's operational models. By integrating this data, we can ensure that Safely's safety heatmap is informed and accurate from the outset, offering users immediate and valuable safety insights.

Integrating User-Generated Data:

As users interact with Safely, we can utilize their input to refine the platform's precision. The data generated by users' reports and feedback will be instrumental in providing a nuanced, real-time portrayal of safety conditions, thus enhancing the platform's adaptability and dependability.

Controlled User Engagement (Alpha Testing):

By inviting a selected group of users to interact with Safely in its early stages, we can closely monitor their experiences and gather critical feedback. This process will allow us to identify and resolve any issues before broader release, ensuring that the app's core functionalities are robust and reliable. Through this meticulous validation of the platform's early versions, we can lay a strong foundation for user safety and a seamless user experience.

Iterative Development Informed by User Feedback (Beta Testing):

We are dedicated to an iterative development approach, with user feedback guiding each cycle of updates and feature rollouts. Through close analysis of user-reported incidents and interactions, we can persistently advance the user experience and amplify the platform's capability in safeguarding users.

Collaborating with Local Authorities:

We can augment the depth and accuracy of our data by collaborating with local law enforcement and safety agencies to integrate real-time crime and safety data. This initiative can yield a richly detailed and current depiction of urban safety landscapes, reinforcing the effectiveness of Safely.

Therefore, through the rigorous testing, we ensure that Safely is perceived as a data centric and user informed platform with the primary goal of having a user-focused, convenient and simple interface which greatly helps in increasing the usability of Safely as an app.

Prototype: User Interface Design and User Experience

Safely's early prototype exhibits a user friendly interface, ensuring ease of access and navigation for users seeking safety information.

Key features showcased in the mockups include:

- **Welcome Screen:** A simple and welcoming sign in screen that quickly re-engages returning users and facilitates new user registration.
- **City Selection:** Users can conveniently select their city, personalizing the safety information and alerts to their specific location.
- **Heatmaps:** The core of Safely's user interface is the interactive safety heatmap. In future iterations, heatmaps will be integrated to provide statistical visualizations of safe and risky areas based on real time data and user reports.
- **SOS Functionality:** A prominently placed SOS button offers users immediate access to emergency services, exemplifying Safely's commitment to user safety.
- **RealTime Tracking:** For premium users, the prototype indicates plans for a 'Start Walk' feature that enables real time tracking to the destination, ensuring users are monitored and assisted throughout their journey.

Works Cited

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